

Promise Dodzi Kpoglu

NLP Research Engineer & Computational Linguist

France · +33 7 68 19 78 56 · promisedodzi@gmail.com · [GitHub](#) · [Website](#)

PROFESSIONAL SUMMARY

Natural Language Processing Research Engineer and Computational Linguist specializing in large-scale language data, multilingual corpora, and data-driven modelling of language evolution. Experienced in building end-to-end computational pipelines for linguistic data processing: from data collection, cleaning, and annotation through to feature engineering and ML model development. My work bridges linguistic theory, statistical modelling, and machine learning, with a focus on unsupervised methods for historical linguistics, cognate detection, and lexical reconstruction.

PROFESSIONAL EXPERIENCE

Research Engineer - Computational Linguistics & NLP

Jan 2024 – Present

LLACAN - CNRS · France

- Lead computational development for linguistic data processing pipelines in multilingual corpora.
- Design and implement machine learning workflows for analysis of linguistic structures and language evolution.
- Develop algorithms for linguistic modelling, feature extraction, and structured dataset generation.
- Responsible for all computational components of the BANG research program (computational historical linguistics).
- Perform data cleaning, normalization, and transformation of raw linguistic datasets into ML-ready formats.

Research Consultant - Linguistic Data Science

May 2016 – Feb 2023

University of Leiden · Netherlands

- Collected, curated, and annotated multilingual linguistic datasets for research project.
- Co-designed data annotation frameworks and quality control procedures for language corpora.
- Conducted analyses of linguistic datasets.
- Built structured datasets for computational comparative linguistics tasks.

Lecturer

Aug 2019 – Dec 2023

University of Ghana · Ghana

- Delivered lectures in linguistics and computational methods.
- Supported research training in data analysis and linguistic methodology.
- Assisted in research activities and lab-based projects.

EDUCATION

PhD in Language Sciences

2014 – 2019

University of Lille / Leiden University · France / Netherlands

Dissertation: [Possessive Constructions in Tongugbe, an Ewe Dialect](#)

Master's Degree in Language Sciences

2012 – 2014

University of Lille · France

SELECTED PROJECTS

Unsupervised Protoform Reconstruction

Developed an unsupervised hybrid model for reconstructing proto-lexical forms from cognate sets.

- Combines statistical inference with rule-based linguistic constraints in an evolutionary optimization framework.
- Evaluated on Romance language datasets; achieved improved character-level accuracy and phonological plausibility vs. baseline models.
- Github: <https://github.com/PromiseDodzi/protoform-reconstruction>

Unsupervised Loanword Detection

Designed an unsupervised system for detecting loanwords using only language-internal signals.

- Extracts linguistic features, assigns probabilistic scores, and iteratively refines classifications without external language metadata.
- Evaluated across English, German, French, Italian, Spanish, and Portuguese datasets.
- Github: https://github.com/PromiseDodzi/loanword_detection

Computational Comparative Linguistics - Dogon Languages

Built an end-to-end pipeline for historical linguistics of Dogon languages.

- Covers preprocessing, segmentation, cognate detection, phonetic alignment, and phylogenetic reconstruction.
- Supports quantitative approaches to language evolution in under-documented language families.
- GitHub: https://github.com/PromiseDodzi/comp_methods_dogon

Linguistic Data Processing Pipeline (BANG Project)

Designed and implemented computational workflows for large-scale linguistic data processing.

- Covers data cleaning, formatting, segmentation, and structuring for downstream modelling and analysis.
- Responsible for all computational aspects of the project.
- Github: https://github.com/IndianaTones/data_merge

Interactive Cognate Detection Tool (Private)

Development of an interactive system for cognate detection and lexical data visualization.

- Integrates linguistic similarity metrics with user-driven exploration.
- Focused on explainability and exploratory historical linguistics workflows.

TECHNICAL SKILLS

Python	SQL	HTML / CSS
PyTorch	Scikit-learn	NumPy / Pandas
Unsupervised Learning	Sequence Modeling	Representation Learning
Linguistic Feature Engineering	Language Model Evaluation	Large-Scale Corpus Construction
Data Cleaning & Normalization	Annotation Pipelines	Multilingual Data Processing
Computational Historical Linguistics	Cognate Detection	Protoform Reconstruction

SELECTED PUBLICATIONS

Journal Articles

- Kpoglu, P.D. (2023). Subject–Verb agreement marking by Ghanaian learners of French. *Southern African Linguistics and Applied Language Studies*.
- Kpoglu, P.D. (2021). Kinship terms and possessive constructions in Tongugbe. *Journal of West African Languages*.
- Kpoglu, P.D. & Boutché, J.P. (2017). Impact of standardisation and vehicularization in African languages. *Histoire Épistémologie Langage*.

Books & Book Chapters

- Kpoglu, P.D. (2019). Possessive Constructions in Tongugbe. PhD Dissertation, LOT.
- Dzahene-Quarshie, J. & Kpoglu, P.D. (eds.) (2023). *Legon Journal of the Humanities*, Special Issue.
- Atintono et al. (2024). Comparative study of locative constructions in African languages. John Benjamins.
- Essizewa et al. (2023). *Togo. Manual of Romance Languages in Africa*.

Conference Proceedings

- Kpoglu & Patin (2018). Depressor consonants and tones in Tonúgbé. *TAL 2018*.

SELECTED TALKS & PRESENTATIONS

- How Dogon Languages Evolve: Insights from Computational Simulation - ACAL 57, University at Buffalo (2026).
- Computational modelling of lexical evolution in Dogon languages - UCAD, Dakar (2024).
- Optimising cognacy for automatic historical linguistics - ENS/PSL, Paris (2024).
- Linguistic history of West African domesticated ungulates - UCAD (2024).
- Lexical segmentation in cognate detection - LATTICE Seminar, ENS Paris (2024).
- Multiple presentations on Ewe syntax, tone systems, and possessive constructions (2016–2021).

LANGUAGES

English (C2) · French (C2) · Ewe (Native) · Twi (Spoken) · Ghanaian Pidgin English (L2)

ADDITIONAL STRENGTHS

- Strong research - engineering integration mindset.
- Experience with multilingual and low-resource language datasets.
- Independent ownership of computational research pipelines.
- Experience in unsupervised modelling approaches.
- Ability to translate linguistic theory into computational systems.